Full-output classical simulation of lossy boson sampling with single-photon and equal-squeezing inputs

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Abstract

We construct classical samplers for two lossy linear-optical models: single-photon Fock inputs at every mode count $m \geq n$, and Gaussian inputs with every mode equally squeezed. At fixed rational transmission $0 < \eta < 1$, and fixed squeezing in the Gaussian model, both algorithms approximate the full photon-number distribution, including all collisions, to Haar-average total variation error ε in worst-case time polynomial in the size parameters and $1/\varepsilon$. The Gaussian result allows a mean surviving photon number proportional to the number of modes. An exact Haar harmonic calculation supplies the truncation bounds: a particle-reduction cancellation for Fock inputs, and a geometric bound obtained from the harmonic spectrum of invariant pair states for equal squeezing. Collision-stratified color coding and finite-ring permanent or hafnian identities evaluate compatible signed marginals in time exponential only in the logarithmic truncation order. We include bounded photon-number draws, matrix precision and random-bit complexity. Further Fock-input corollaries give explicit loss–accuracy dependence and an orthogonal bad-bit distinguishability extension. Small-system Haar experiments and exact algebraic checks accompany the proofs. The algorithms are asymptotic constructions; no practical runtime advantage is claimed.

This manuscript has undergone internal proof review and computational checks. Independent expert review remains outstanding; Section 11 describes the evidence and AI assistance.

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1 Two input models and their simulation guarantees

Boson sampling asks a classical computer to reproduce the photon-count statistics of a passive linear-optical experiment [1]. Photon loss suppresses many-particle interference, suggesting that only low orders need to be retained. We turn this truncation into a full-output sampler in two dense regimes: single-photon inputs with as few modes as photons, and equally squeezed Gaussian inputs occupying every mode. The Gaussian calculation requires its own input spectrum; it does not follow by substituting a mean photon number into the Fock bound.

Inject one indistinguishable photon into each of the first n inputs of an m-mode interferometer, with $m \geq n \geq 1$. Each photon survives independently with probability η. For $U \in U(m)$ and an occupation $s \in \mathbb{N}_0^m$, write $|s| = \sum_i s_i$ and $s! = \prod_i s_i!$. The target law is

$$p_{\eta,U}(s) = \sum_{\substack{I \subseteq [n] \\ |I|=|s|}} \eta^{|I|}(1-\eta)^{n-|I|} \frac{|\operatorname{Per}(U_{s,I})|^2}{s!}. \tag{1.1}$$

Here $U_{s,I}$ repeats row i exactly s_i times and selects the columns in I. The empty permanent is one. In particular the output includes every collision pattern, together with the vacuum event. Total variation is $\mathrm{TV}(p,q) = \frac{1}{2}\sum_s |p(s) - q(s)|$.

Theorem 1.1 (Single-photon inputs)**.** *Fix rational $0 < \eta < 1$. There is a randomized classical algorithm with the following property. For every $m \geq n \geq 1$ and rational accuracy $0 < \varepsilon < 1$, an entrywise encoding of U using $\mathrm{poly}_\eta(n, m, \log(1/\varepsilon))$ bits suffices to produce an output law q_U satisfying*

$$\mathbb{E}_{U \sim \mathrm{Haar}(U(m))} \mathrm{TV}(q_U, p_{\eta,U}) \leq \varepsilon. \tag{1.2}$$

The contribution of matrix encoding is at most $\varepsilon/2$. The worst-case running time, including the generation of random bits, is polynomial in $n, m, 1/\varepsilon$ and the binary input length. The polynomial can depend on fixed η; its exponent does not depend on ε. Proposition 8.1 makes the dependence uniform when η is an input parameter, and Proposition 9.1 treats a specified partial-distinguishability model.

Theorem 1.2 (Every mode equally squeezed)**.** *Fix rational $0 < \zeta, \eta < 1$. Prepare all m modes in the same squeezed vacuum, with squared hyperbolic tangent of the squeezing parameter equal to ζ. Apply uniform pure loss of transmission η, a Haar-random passive interferometer $U \in U(m)$, and photon-number-resolving measurement on every mode. Write $p^{\mathrm{G}}_{\zeta,\eta,U}$ for the full output law. There is a randomized classical algorithm whose law q^{G}_U satisfies*

$$\mathbb{E}_U \mathrm{TV}(q^{\mathrm{G}}_U, p^{\mathrm{G}}_{\zeta,\eta,U}) \leq \varepsilon. \tag{1.3}$$

An entrywise encoding of U with $\mathrm{poly}_{\zeta,\eta}(m, \log(1/\varepsilon))$ bits suffices. The worst-case bit complexity is polynomial in $m, 1/\varepsilon$ and the binary input length, with degree depending on fixed ζ, η. The output includes vacuum, every total photon number and every collision pattern.

The mean number of detected photons in Theorem 1.2 is $m\eta\zeta/(1-\zeta)$, hence linear in m at fixed parameters. Its equal-squeezing assumption applies to every mode: it does not cover additional vacuum inputs or arbitrary squeezing profiles. The two theorems concern different input families. Both use Haar-average error, and neither asserts a guarantee for each deterministic interferometer. The loss–accuracy refinement in Section 8 applies to Theorem 1.1; we do not assert a corresponding uniform limit $\eta \to 1$ for Theorem 1.2.

Theorem 1.1 addresses the question posed in the Quantum Information and Quantum Computation Open Problem Zoo [14]. The error is averaged over U, with random preprocessing included in q_U. The dependence on fixed η becomes expensive as η approaches one; Section 4.4 gives an explicit conservative bound. The special case $m = n = 1$ is a single Bernoulli draw; the Gaussian $m = 1$ case likewise reduces to bounded photon-number draws. The harmonic arguments below use $m \geq 2$.

1.1 Prior work and the remaining mode-density restriction

Low-degree approximation under noise has a substantial history. Kalai and Kindler [2] studied Gaussian noise in permanent probabilities. Renema, Shchesnovich and García-Patrón [3] organized noisy boson-sampling probabilities by interference order, including the effect of photon loss. Shchesnovich [4] related Gaussian noise to a model combining loss and dark counts. Other efficient simulations use stronger loss [5] or depth-dependent attenuation in optical architectures [6]. These works provide distinct noise models and simulation regimes; the model here keeps a constant fraction of the input photons on average.

The closest full-output comparison is Go and Jeong [7]. Write N for their input photon count, k for the fixed survivor count and M for their mode count. Their Corollary 4 assumes $M \geq C_0 k^2 N$ and $M \geq C_B k^2/(\varepsilon\delta)$, and gives TV error at most ε with Haar probability at least $1-\delta$. Their Section XI identifies full-output simulation at comparable mode and photon counts as an open direction. Theorem 1.1 removes these mode restrictions. Its average-error formulation also gives a high-probability statement: running at mean tolerance $\varepsilon\delta$ bounds the probability of TV error exceeding ε by δ, with polynomial dependence on $1/\delta$. Thus the substantive scope difference is the allowed mode density. The additional birthday bound in [7] controls collisions, whose onset is studied in the bosonic birthday paradox [8].

Exact Haar analysis itself is established prior work. Nezami [9] uses representation theory to calculate moments of permanents of unitary submatrices. Arienzo et al. [10] provide the passive-bosonic operator decomposition and photon-number measurement frame used below. Mhiri et al. [11], Proposition 2 and Corollary 1, express Haar second moments through irreducible projection norms and derive anti-concentration results beyond the dilute regime. We use this sectorwise viewpoint to control loss, then construct the collision-resolving marginal evaluator needed for sampling.

1.2 Gaussian simulation and the role of averaging

Gaussian boson sampling expresses photon probabilities through hafnians [19]; exact exponential-time samplers are available [20]. Approximation algorithms have exploited loss, distinguishability and low-order correlations. Renema [22] studies imperfect Gaussian and superposition inputs, with a remaining permanent-approximation condition. Umanskii and Rubtsov [23] approximate probabilities by a Taylor expansion whose fixed-order cost is polynomial in mode count. Oh et al. [24] give a tensor-network simulation with strong empirical performance and a provable high-loss regime. Dodd et al. [26] give a practical cumulant-based emulator for threshold detection, while Liu et al. [27] accelerate MPS construction for low-entanglement Gaussian states. These results address important computational settings; fixed-order probability evaluation or experimental benchmark agreement alone does not supply the full-output, inverse-accuracy guarantee in (1.3).

The closest recent loss comparison is Lim [18]. That work proves an all-interferometer TV bound $O(N\eta^3)$ at polynomial expected cost, permitting $N^{2/3}$ mean surviving photons at fixed accuracy, and a separate pair construction reaches $N^{3/4}$ when every mode is equally squeezed. Theorem 1.2 instead keeps η fixed and permits a linear mean output size, using Haar averaging and a worst-case bit bound. The quantifiers differ: our theorem does not improve Lim's guarantee for every interferometer. The Gaussian moment and hafnian generating identities used in our oracle are established tools [19, 21]; the contribution here is their combination with an exact pair-state harmonic bound and a collision-resolving marginal evaluator.

Hardness results for weakly lossy Gaussian sampling [25] address a different scaling toward unit transmission. Our fixed-parameter theorem does not locate a sharp hardness transition. Likewise, the Fock corollary for vanishing loss below leaves a gap from known low-loss hardness regimes [15, 16].

1.3 Construction and notation

Trace-one particle reduction produces a factorial cancellation, after which independent loss suppresses harmonic degree l by the exact factor η^l. Collision patterns are grouped by their modes occupied at least twice, while singleton modes are color coded. Color coding originates with Alon, Yuster and Zwick [12]; its use with inverse determinants for this sampling problem follows the framework of [7]. The evaluator preserves both harmonic projections and has cost $\mathrm{poly}(n,m)\exp(O(r))$. A signed-prefix construction [13, 7] then turns its compatible marginals into a probability sampler. For the Fock model, the extension is this combination of the loss estimate and the collision-resolving, two-projection evaluator at arbitrary mode density.

Let $H_k = \mathrm{Sym}^k \mathbb{C}^m$, $D_k = \binom{m+k-1}{k}$ and $D_{-1} = 0$. Let $\Gamma_k(U)$ be the passive action on H_k. The trace-one j-particle reduction is $R_{k,j}$, and its Hilbert–Schmidt adjoint is $L_{k,j} = R_{k,j}^*$. Falling and rising factorials are written $(a)_l$ and $a^{\overline{l}}$. The input conditional on $K = k$ surviving photons is

$$\rho_{k,n} = \binom{n}{k}^{-1} \sum_{\substack{I\subseteq[n]\\|I|=k}} |I\rangle\langle I|, \qquad w_k = \Pr(K=k) = \binom{n}{k}\eta^k(1-\eta)^{n-k}. \tag{1.4}$$

It obeys $R_{k,j}\rho_{k,n} = \rho_{j,n}$ and $\mathrm{Tr}\,\rho_{j,n}^2 = \binom{n}{j}^{-1}$. We write $p_{U,k}$ for its output Fock diagonal. Thus (1.1) is the mixture $w_k p_{U,k}$ on the disjoint photon-number sectors.

2 An exact Haar harmonic error bound

2.1 Representation and measurement normalization

The conjugation representation on $\mathrm{End}(H_k)$ is an orthogonal multiplicity-free sum of irreducible sectors $V_{k,l}$, $0 \le l \le k$, with highest weight $(l, 0, \dots, 0, -l)$ and dimension

$$d_l = D_l^2 - D_{l-1}^2. \tag{2.1}$$

Let $\Pi_{k,l}$ be the corresponding orthogonal projector. These representation facts are imported from [10], Lemma T1 and Proposition B2. The same source, Lemma T3 and Theorem T4, identifies the photon-number measurement frame. In our normalization its eigenvalue on sector l is

$$f_l = \frac{1}{D_l + D_{l-1}}, \qquad f_0 = 1. \tag{2.2}$$

Indeed Fock dephasing $\mathcal{D}_k$ is averaging over the diagonal unitary torus. Its rank on $V_{k,l}$ is the zero-weight multiplicity $D_l - D_{l-1}$; dividing this trace by d_l gives (2.2). Equivalently, the source expression is

$$\frac{m-1}{(2l+m-1)\binom{l+m-2}{l}} = f_l.$$

For $X \in V_{k,l}$ this implies

$$\mathbb{E}_U \|\mathcal{D}_k(\Gamma_k(U) X \Gamma_k(U)^*)\|_{\mathrm{HS}}^2 = f_l \|X\|_{\mathrm{HS}}^2. \tag{2.3}$$

Dephasing commutes with sector projection, since the latter commutes with every conjugation action. Consequently diagonal vectors in different sectors remain orthogonal, even before Haar averaging.

2.2 Particle reduction

Lemma 2.1 (Reduction singular value)**.** *Put*

$$B(k,l) = (l!)^2 \binom{m+k+l-1}{k-l}. \tag{2.4}$$

For $l \le j \le k$, the squared Hilbert–Schmidt singular value of $R_{k,j}$ on $V_{k,l}$ is

$$t_{k,j,l} = \left(\frac{(j)_l}{(k)_l}\right)^2 \frac{B(k,l)}{B(j,l)}. \tag{2.5}$$

Reduction annihilates sectors $l > j$.

Proof. Use $A_{k,l} = (a_1^\dagger)^l a_m^l$ on H_k. This is a highest-weight vector in $V_{k,l}$. The squared transition coefficient at occupation s is $(s_1+l)!/s_1!$ times $s_m!/(s_m-l)!$. Its generating function in total photon number is

$$\frac{l!}{(1-z)^{l+1}} \frac{l!z^l}{(1-z)^{l+1}} \frac{1}{(1-z)^{m-2}} = \frac{(l!)^2 z^l}{(1-z)^{m+2l}}.$$

Coefficient extraction gives $\|A_{k,l}\|_{\mathrm{HS}}^2 = B(k,l)$. The trace-one partial trace satisfies

$$R_{k,j}^*(A_{j,l}) = \frac{(j)_l}{(k)_l} A_{k,l}.$$

This follows by expressing the normally ordered l-particle observable as a sum over ordered distinct particle labels. Passing to a uniformly selected j-particle subsystem retains the displayed fraction of these labels. Equivariance and multiplicity-freeness make the map scalar between normalized highest weights. Their norms give (2.5). There is no copy of sector $l > j$ in the target. □

For $W \in V_{l,l}$ define the compatible lift

$$J_{k,l} W = \frac{L_{k,l} W}{t_{k,l,l}} \quad (k \ge l). \tag{2.6}$$

It has squared norm $\|W\|_{\mathrm{HS}}^2 / t_{k,l,l}$ and l-particle reduction W. Transitivity of reduction and uniqueness in the irreducible sector give

$$R_{k,j} J_{k,l} W = J_{j,l} W \ (j \ge l), \qquad R_{k,j} J_{k,l} W = 0 \ (j < l). \tag{2.7}$$

Lifts of diagonal operators are diagonal. Each $l > 0$ lift has trace zero.

2.3 Factorial cancellation and loss averaging

Lemma 2.2 (Dimension-uniform cancellation)**.** *For $1 \le l \le k \le n \le m$,*

$$\frac{D_k f_l}{t_{k,l,l} \binom{n}{l}} = \frac{(k)_l}{(n)_l} \frac{(m+l)^{\bar{l}}}{(m+k)^{\bar{l}}} \frac{m+l-1}{m+2l-1} \le \frac{(k)_l}{(n)_l}. \tag{2.8}$$

Proof. Substitute (2.5) and (2.2), then use

$$\frac{D_k}{\binom{m+k+l-1}{k-l}} = \frac{m^{\overline{2l}}}{(k)_l (m+k)^{\bar{l}}}, \qquad D_l + D_{l-1} = \frac{m^{\bar{l}}}{l!} \frac{m+2l-1}{m+l-1}.$$

Both remaining ratios in (2.8) are at most one. □

Write $\rho_l^{(k)} = \Pi_{k,l}\rho_{k,n}$. Its reduction lies in the lth sector of $\rho_{l,n}$, so

$$t_{k,l,l}\|\rho_l^{(k)}\|_{\mathrm{HS}}^2 \le \binom{n}{l}^{-1}. \tag{2.9}$$

This is an upper bound, not an assertion that each sector saturates it. Define the normalized signed approximation

$$q_{U,k,r}(s) = \langle s|\Gamma_k(U) \sum_{l=0}^{\min(r,k)} \rho_l^{(k)}\Gamma_k(U)^*|s\rangle.$$

Equations (2.3)–(2.9) imply

$$\mathbb{E}_U D_k \sum_{|s|=k} |p_{U,k}(s) - q_{U,k,r}(s)|^2 \le \sum_{l=r+1}^{k} \frac{(k)_l}{(n)_l}. \tag{2.10}$$

To pass to loss, use the probability reference w_k/D_k in Cauchy–Schwarz and then Jensen's inequality. Since $\mathbb{E}(K)_l/(n)_l = \eta^l$ exactly,

$$\begin{aligned}\mathbb{E}_U\|p_{\eta,U} - q_{U,r}\|_1 &\le \left(\sum_k w_k D_k \mathbb{E}_U \sum_{|s|=k} |p_{U,k}(s) - q_{U,k,r}(s)|^2\right)^{1/2} \\ &\le \left(\sum_{l=r+1}^{n} \eta^l\right)^{1/2} \le \frac{\eta^{(r+1)/2}}{\sqrt{1-\eta}},\end{aligned} \tag{2.11}$$

where $q_{U,r}(s) = w_{|s|}q_{U,|s|,r}(s)$. The tail is zero for $r = n$. This argument includes $K = n$ and every colliding output; it does not remove rare survivor counts or replace Haar matrices by Gaussian matrices.

3 A collision cutoff and one consistent random signed law

3.1 Counting the heavy supports

For an l-photon occupation v, define its deficit $d(v) = l - |\operatorname{supp}(v)|$ and heavy support $H(v) = \{i : v_i \ge 2\}$. Write $b_i = v_i$ on $H(v)$ and $s = l - \sum_{i\in H(v)} b_i$. Thus s singleton modes remain and $d(v) = \sum_{i\in H(v)}(b_i - 1)$.

The uniform law on the D_l occupations has a Polya-urn realization: start with one ball of each of m colors and make l additions. After a additions, the probability of selecting an occupied color is at most $2a/m$. Every such repeat increases the deficit by one. A union bound over D selected addition times, using $\binom{l}{D} \le (el/D)^D$, gives

$$\Pr_{v\,\text{uniform}}[d(v) \ge D] \le \min\left\{1, \left(\frac{6l^2}{mD}\right)^D\right\} \quad (1 \le D < l). \tag{3.1}$$

For $D = l$ the probability is zero. Given $0 < \tau < 1$, let $D(l)$ be the first $D \in \{1, \dots, l\}$ for which this upper bound, with the $D = l$ value set to zero, is at most τ. Retain exactly the occupations with $d(v) < D(l)$.

Lemma 3.1 (Efficient heavy-support enumeration)**.** *With $C = 4$,*

$$\sum_{q=0}^{\min(D(l)-1,\lfloor l/2\rfloor)} \binom{m}{q} \le (l+1)\tau^{-1}e^{Cl}. \tag{3.2}$$

For each support, its possible heavy occupancies can be enumerated in $\mathrm{poly}(l)2^l$ *time.*

Proof. For $D(l) = 1$ only the empty support occurs. Otherwise $d = D(l) - 1$ failed the cutoff test, so $d \log(md/(6l^2)) < \log(1/\tau)$. For $1 \le q \le d \le l \le m$, monotonicity of $x \log(em/x)$ gives

$$\begin{aligned}\log \binom{m}{q} &\le q \log(em/q) \le d \log(em/d) \\ &< \log(1/\tau) + d \log(6e) + 2d \log(l/d) \le \log(1/\tau) + Cl.\end{aligned}$$

Here $d \le l$ and $d \log(l/d) \le l/e$, so $C = \log(6e) + 2/e < 4$ suffices. Sum over at most $l + 1$ support sizes. For a support of size q, recursively enumerate $b_i \ge 2$ with $\sum_i b_i \le l$, restricting the next value so that the remaining positions can each receive at least two. The number of assignments is bounded by the number of compositions of totals at most l, hence by 2^l; the tree overhead is polynomial in l. Apply the deficit test to these assignments. □

The cutoff concerns occupations of the low harmonic degree l, not the physical survivor number k. It does not declare all physical output collisions rare or replace them by a common symbol.

3.2 Input and output colors

Following the color-coding method [12], for each $1 \le l \le r$ choose independently a uniform input coloring $h_l : [n] \to [l]$. Set $p_l = l!/l^l$ and define

$$X_l = \frac{1}{p_l \binom{n}{l}} \sum_{\substack{I \subseteq [n],\ |I|=l \\ h_l(I)=[l]}} |I\rangle\langle I|, \qquad W_l = \Pi_{l,l} X_l. \tag{3.3}$$

The trace of X_l can fluctuate. Its expectation is $\rho_{l,n}$, and orthogonality of the diagonal terms gives

$$\mathbb{E}\|W_l\|_{\mathrm{HS}}^2 \le \mathbb{E}\|X_l\|_{\mathrm{HS}}^2 = \frac{1}{p_l \binom{n}{l}}. \tag{3.4}$$

Every W_l is traceless.

For each $s \in \{1, \dots, l\}$ choose a uniform output coloring $g_{l,s} : [m] \to [s]$. Let $S(v)$ be the singleton support of v and $s = |S(v)|$. Define $p_0 = 1$ and

$$a_l(v) = \mathbf{1}_{d(v)<D(l)} \frac{\mathbf{1}_{g_{l,s}(S(v))=[s]}}{p_s}. \tag{3.5}$$

The empty singleton condition is automatically satisfied. Let $\mathcal{D}_{a_l}$ discard off-diagonal entries and multiply the diagonal at v by $a_l(v)$. Its mean is the projection $\mathcal{D}_{\mathrm{good}}$ onto retained diagonal labels, and $\mathbb{E}_g a_l(v)^2 \le e^l$.

For one collection of all colors set

$$A_l(U) = \Pi_{l,l} \mathcal{D}_{a_l}\big(\Gamma_l(U) W_l \Gamma_l(U)^*\big), \qquad B_k(U) = \frac{I}{D_k} + \sum_{l=1}^{\min(r,k)} J_{k,l} A_l(U). \tag{3.6}$$

The outer projection places A_l in $V_{l,l}$, so (2.7) applies. The matrices B_k are diagonal, Hermitian, trace one and particle compatible. Average T independent complete color collections, keep them fixed throughout a sample, and denote the resulting matrices by $\overline{B}_k$. Then

$$\widetilde{q}_U(s) = w_{|s|} \langle s | \overline{B}_{|s|}(U) | s \rangle$$

is a single normalized signed law. Reusing the same colors for each prefix preserves marginal consistency.

3.3 Bias, variance and the total error

For a fixed $W \in V_{l,l}$, Schur orthogonality yields

$$\mathbb{E}_U |\langle v|\Gamma_l(U) W \Gamma_l(U)^* |v\rangle|^2 = \frac{\|W\|_{\mathrm{HS}}^2 \|\Pi_{l,l}(|v\rangle\langle v|)\|_{\mathrm{HS}}^2}{d_l} \le \frac{\|W\|_{\mathrm{HS}}^2}{d_l}. \tag{3.7}$$

At most τD_l labels are discarded. Moreover,

$$\frac{D_l}{d_l} = \frac{D_l}{D_l - D_{l-1}} f_l = \frac{m+l-1}{m-1} f_l \le 3 f_l,$$

because $m \ge l$ and $m \ge 2$. In the mean of (3.6), use the actual input $W = \Pi_{l,l}\rho_{l,n}$, whose squared norm is at most $\binom{n}{l}^{-1}$. Projection contraction, the lift norm, and (2.8) bound the D_k-weighted squared bias at degree l by

$$3\tau \frac{(k)_l}{(n)_l}. \tag{3.8}$$

Here the unmasked target is the exact dephased harmonic component: the equivariant lift commutes with conjugation and with torus dephasing.

For the variance, condition on X_l and U. Equation (3.5) and projection contraction imply

$$\mathbb{E}_g \|A_l(U)\|_{\mathrm{HS}}^2 \le e^l \|\mathcal{D}_l(\Gamma_l(U) W_l \Gamma_l(U)^*)\|_{\mathrm{HS}}^2.$$

Average over U and then h_l. Since $p_l^{-1} \le e^l$, equations (2.3) and (3.4) give $\mathbb{E}\|A_l(U)\|_{\mathrm{HS}}^2 \le e^{2l} f_l / \binom{n}{l}$. Centered variance is no larger. Independence of the T collections and (2.8) therefore bound the weighted lifted variance by

$$T^{-1} e^{2l} \frac{(k)_l}{(n)_l}. \tag{3.9}$$

Errors at different degrees are orthogonal in diagonal Hilbert–Schmidt space. Applying the same weighted Cauchy–Schwarz argument as in (2.11), separately to bias and centered fluctuation, gives

$$\mathbb{E}_{U,\mathrm{colors}} \|p_{\eta,U} - \widetilde{q}_U\|_1 \le \frac{\eta^{(r+1)/2}}{\sqrt{1-\eta}} + \sqrt{\frac{3\tau}{1-\eta}} + \left(T^{-1} \sum_{l=1}^{r} e^{2l} \eta^l \right)^{1/2}. \tag{3.10}$$

Omit the first term when $r = n$. Correlations among the output-color indicators are included: the argument does not assume independent monomials.

Select r as the minimum of n and the least nonnegative integer for which $\eta^r \le \varepsilon^2(1-\eta)/64$, and put

$$\tau = \frac{\varepsilon^2(1-\eta)}{192}, \qquad T = \left\lceil 64 \varepsilon^{-2} r 9^r \right\rceil. \tag{3.11}$$

Here $r \ge 1$. Each term in (3.10) is at most $\varepsilon/8$, hence the total is at most $3\varepsilon/8$. Parameter selection uses rational comparisons. For fixed η, $r = O_\eta(1 + \log(1/\varepsilon))$, while T and τ^{-1} are polynomial in $1/\varepsilon$.

4 Evaluating prefixes while retaining collisions

4.1 Both projections in reduced-particle coordinates

On H_l, compute rational coefficients c_a by

$$c_l = 1, \qquad c_j = -\frac{\sum_{a=j+1}^{l} c_a t_{l,a,j}}{t_{l,j,j}} \quad (j = l-1, \dots, 0). \tag{4.1}$$

The action on each irreducible sector gives the exact identity

$$\Pi_{l,l} = \sum_{a=0}^{l} c_a L_{l,a} R_{l,a}. \tag{4.2}$$

These equations invert operator maps; they do not divide by a possibly vanishing state component.

Fix l, suppress the index on its input coloring, and write $N_c = |h_l^{-1}(c)|$. For a colorful a-subset I,

$$\langle I | R_{l,a} X_l | I \rangle = \frac{\prod_{c \notin h_l(I)} N_c}{p_l \binom{n}{l} \binom{l}{a}}. \tag{4.3}$$

All other diagonal entries vanish. This simply counts colorful l-subsets extending I. The lift of a diagonal matrix satisfies the hypergeometric formula

$$\langle v | L_{l,a}(Z) | v \rangle = \binom{l}{a}^{-1} \sum_{\substack{u \le v \\ |u|=a}} \binom{v}{u} \langle u | Z | u \rangle, \qquad \binom{v}{u} = \prod_i \binom{v_i}{u_i}. \tag{4.4}$$

Equivariance, (4.3) and (4.4) give

$$\begin{aligned} Z_l(v) &:= \langle v | \Gamma_l(U) \Pi_{l,l} X_l \Gamma_l(U)^* | v \rangle \\ &= \sum_{a=0}^{l} \frac{c_a}{p_l \binom{n}{l} \binom{l}{a}^2} \sum_{\substack{I \subseteq [n],\ |I|=a \\ I \text{ colorful}}} \left(\prod_{c \notin h_l(I)} N_c \right) \sum_{\substack{u \le v \\ |u|=a}} \binom{v}{u} \frac{|\operatorname{Per}(U_{u,I})|^2}{u!}. \end{aligned} \tag{4.5}$$

One factor $\binom{l}{a}$ comes from input reduction and the other from lifting.

Represent a k-photon outcome by a uniformly shuffled ordered list of its photons. A particular prefix of length ℓ has occupation h, with $\sum_i h_i = \ell$. Its probability under a compatible family is $h!/\ell!$ times the hth Fock diagonal of the ℓ-particle matrix. Degrees $l > \ell$ contribute zero. For $l \le \ell$, moving the outer projector in (3.6) onto the prefix test gives

$$\begin{aligned} F_{b,h}(x) &= \sum_{t=0}^{\min(b,h)} \binom{b}{t} \binom{h}{t} x^t, \\ K_{\ell,l}(h, v) &= \sum_{j=0}^{l} \frac{c_j}{\binom{l}{j} \binom{\ell}{j}} [x^j] \prod_i F_{v_i,h_i}(x). \end{aligned} \tag{4.6}$$

To see this, apply (4.2) to $R_{\ell,l}|h\rangle\langle h|$, use reduction transitivity, and then (4.4). The convolution of the two hypergeometric weights is exactly the coefficient in (4.6). Thus the degree-l contribution to the ordered prefix is

$$\frac{h!}{\ell!\, t_{\ell,l,l}} \sum_{|v|=l} a_l(v) Z_l(v) K_{\ell,l}(h, v). \tag{4.7}$$

The scalar reference contributes $h!/(\ell! D_\ell)$. Summing degrees and averaging the fixed T collections gives the required prefix marginal.

4.2 A finite-ring coefficient identity

Fix an enumerated heavy support H and its occupancies $b_i \ge 2$. Put $s = l - \sum_{i \in H} b_i$. Introduce commuting square-zero variables $X_1, \dots, X_s$ and $Y_1, \dots, Y_l$, heavy variables z_i truncated above

degree b_i, and x truncated above degree l. These operations take place in a finite commutative quotient ring over Gaussian rationals.

Let V be the first n columns of U, $Y = \operatorname{diag}(Y_{h_l(1)}, \dots, Y_{h_l(n)})$, and define

$$f_i(x) = 1 + h_i x, \qquad Z_i = \begin{cases} X_{g_{l,s}(i)} f_i(x), & i \notin H,\ s > 0, \\ 0, & i \notin H,\ s = 0, \\ z_i, & i \in H. \end{cases}$$

Compute

$$G = \left[\prod_{i \notin H} (1 + X_{g_{l,s}(i)} f_i(x)) \right] \det(I_m - \operatorname{diag}(Z) V Y V^*)^{-1} \prod_{i \in H} F_{b_i,h_i}(x), \tag{4.8}$$

where the first product is one when $s = 0$. It can be grouped by color since each $X_c^2 = 0$.

For each $T_0 \subseteq [l]$, $0 \le u_i \le b_i$ on H, and $0 \le j \le l$, extract the coefficient of

$$X_1 \cdots X_s \prod_{c \in T_0} Y_c \prod_{i \in H} z_i^{u_i} x^j.$$

Multiply it by

$$\frac{c_{|T_0|} c_j \prod_{c \notin T_0} N_c \prod_{i \in H} \binom{b_i}{u_i}}{p_l \binom{n}{l} \binom{l}{|T_0|}^2 \binom{l}{j} \binom{\ell}{j} p_s}. \tag{4.9}$$

Sum over coefficients and all retained heavy patterns, then multiply by $h!/(\ell! t_{\ell,l,l})$. The result is (4.7).

Proof of the coefficient identity. MacMahon's inverse-determinant identity reads

$$\det(I - \operatorname{diag}(Z) C)^{-1} = \sum_u \frac{\operatorname{Per}(C_u)}{u!} Z^u,$$

where C_u repeats both rows and columns. Put $C = VYV^*$ and expand by permanent Cauchy–Binet. A squarefree color coefficient selects distinct input modes with precisely those colors; its value is the sum of $|\operatorname{Per}(V_{u,I})|^2/u!$ over those modes.

A singleton mode selected actively in the determinant contributes one X and one $f_i(x)$. If it is inactive in u, the external product in (4.8) can instead select it as a spectator in v. Selecting the same color twice vanishes, so an active singleton cannot also be counted as a spectator, and distinct singleton modes cannot share a retained color. Consequently extraction of $X_1 \cdots X_s$ selects exactly the colorful singleton supports, with one $f_i(x)$ per selected mode. Heavy modes are fixed by H, b and may participate u_i times; their binomial weights are inserted by (4.9). The last product in (4.8) supplies their factors in (4.6). The remaining weights are precisely (4.5) and (4.6). This includes empty input-color classes and $s = 0$. □

4.3 Arithmetic complexity

The ring has at most

$$(l+1) 2^{s+l} \prod_{i \in H} (b_i + 1) \le (l+1) 4^l \tag{4.10}$$

coefficients, because $b + 1 \le 2^b$ and $s + \sum_i b_i = l$. Schoolbook ring multiplication costs $\exp(O(l))$ scalar operations. Sylvester's determinant identity reduces (4.8) to $\det(I_n - M)^{-1}$

with $M = YV^* \operatorname{diag}(Z)V$. Every entry of M has positive Y-degree, so $M^{l+1} = 0$ in this ring. Evaluate

$$\det(I_n - M)^{-1} = \exp\left(\sum_{a=1}^{l} \frac{\operatorname{Tr}(M^a)}{a}\right), \tag{4.11}$$

truncating the exponential at degree l. This takes $\operatorname{poly}(n, m, l)\exp(O(l))$ operations, with neither an $n!$ determinant expansion nor enumeration of n^l input subsets.

Lemma 3.1, all degrees, T collections, and at most nm prefix queries yield the arithmetic bound

$$\operatorname{poly}(n, m, r)\, T\, \tau^{-1} \exp(O(r)). \tag{4.12}$$

Since $r = O_\eta(1 + \log(1/\varepsilon))$, this is polynomial in the required size parameters for each fixed η. Section 5 supplies the bit complexity and the sampler.

4.4 Dependence on transmission and accuracy

The dependence on accuracy can be made explicit, although the following upper bound is deliberately conservative. Put $a_\eta = \log(1/\eta)$. The choice in (3.11) satisfies

$$r \le \left\lceil \frac{\log(64/((1-\eta)\varepsilon^2))}{a_\eta} \right\rceil.$$

The number of color collections alone is therefore

$$T = \widetilde{O}_\eta(\varepsilon^{-\beta(\eta)}), \qquad \beta(\eta) = 2 + \frac{2\log 9}{a_\eta}.$$

This exponent omits the work within each collection. To bound that work, use $C = 4$ in Lemma 3.1. The heavy occupancy assignments contribute 2^l, and schoolbook multiplication in the ring of (4.10) contributes at most $\operatorname{poly}(l)16^l$. Thus (4.12) can be bounded explicitly by

$$\operatorname{poly}(n, m, r)\, T\tau^{-1}(32e^4)^r \le \operatorname{poly}(n, m, r)\, O_\eta\left(\varepsilon^{-4}(288e^4)^r\right). \tag{4.13}$$

For fixed η, a conservative arithmetic-operation count is consequently

$$\operatorname{poly}(n, m)\, \widetilde{O}_\eta(\varepsilon^{-\alpha(\eta)}), \qquad \alpha(\eta) = 4 + \frac{2\log(288e^4)}{a_\eta}. \tag{4.14}$$

The tilde hides powers of $\log(1/\varepsilon)$. These are upper bounds for the specified construction, not optimized exponents or measured runtimes.

Transmission η	Color count $\beta(\eta)$	Full arithmetic bound $\alpha(\eta)$
0.25	5.17	17.94
0.50	8.34	31.88
0.75	17.28	71.18
0.90	43.71	187.43

In particular, a small exponent for the number of color collections does not give the exponent of the whole algorithm. These bounds establish polynomial scaling for fixed transmission, while leaving practical efficiency unresolved.

5 A proper sampler with polynomial bit complexity

5.1 Sampling a normalized signed law

Sampling from approximately correct signed marginals appears in Bremner, Montanaro and Shepherd [13], Section 3.2 and Appendix B, for binary strings, and in [7], Lemma 8. Here the signed law has mass exactly one. We prove the resulting finite-alphabet bound and its dependence on marginal precision.

Lemma 5.1 (Clipped-prefix sampling)**.** *Let q be a real signed function of total mass one on length-k strings, and suppose all prefix marginals can be queried. At a prefix v, choose child w proportionally to $q(w)_+ = \max(q(w), 0)$. This defines a probability law S and, for every probability law p,*

$$S(v) \le q(v)_+ \quad \text{at every prefix}, \qquad \mathrm{TV}(S, p) \le \tfrac{1}{2}\|q - p\|_1. \tag{5.1}$$

If each queried child marginal is replaced by a value within δ of its true value, the change in the sampled law is at most $2km^k\delta$ for an alphabet of size m. A fallback distribution may be used if all approximate positive weights vanish.

Proof. The empty-prefix marginal equals one. Write $C(v) = \sum_{w \text{ child of } v} q(w)_+ \ge q(v)$. Induction gives $S(v) \le q(v)_+$ and, at any reachable prefix, $q(v) > 0$ and $C(v) > 0$. Then $S(w) = S(v)q(w)_+/C(v) \le q(w)_+$. At leaves,

$$\mathrm{TV}(S, p) = \sum_x (S(x) - p(x))_+ \le \sum_x (q(x) - p(x))_+ = \tfrac{1}{2}\|q - p\|_1.$$

Clipping changes each child weight by at most δ. The conditional distributions before and after approximation differ in TV by at most $2m\delta/C(v)$, capped at one. If all approximate weights vanish, $C(v) \le m\delta$, so the same capped estimate covers an arbitrary fallback. Couple paths until their first discrepancy. Weighting a node by the probability $S(v) \le C(v)$ bounds its contribution by $2m\delta$. At most km^{k-1} nonterminal prefixes exist, which proves the stated conservative bound. The sampler visits only one path and makes at most km oracle queries; the full tree is used only in the error estimate. □

Apply the lemma conditional on $K = k$ to the ordered law of $\overline{B}_k$, with all colors fixed. Forgetting order cannot increase TV. Averaging over K, colors and U, convexity and (3.10) give an ideal-sampling error at most $3\varepsilon/8$, even using the weaker bound $\mathrm{TV} \le \|q - p\|_1$. This conservative allocation leaves room for encoding and random-bit approximation.

5.2 Uniform matrix precision

The prefix identities in Section 4 are polynomials in entries of U and their conjugates of degree at most $2r$. They remain explicit polynomials when evaluated at a rounded matrix, even though that matrix need not be unitary. We bound their perturbation directly; no Haar theorem for rounded matrices is assumed.

One deliberately loose coefficient bound is as follows:

$$\begin{aligned} M_0 &= 4^r(m + 2r)^r, & C_0 &= (1 + rM_0)^r, \\ B_0 &= 1 + (r + 1)^6 C_0^2 (r!)^2 \left[64(m + 2r)(n + 1)^6\right]^r. \end{aligned} \tag{5.2}$$

Every top-projector coefficient for $l \le r$ satisfies $|c_j| \le C_0$. Indeed $t_{l,a,j} \le (m + 2r)^r$ and $t_{l,j,j} \ge \binom{l}{j}^{-2} \ge 4^{-r}$, and backward substitution in (4.1) gives this bound.

For each fixed color collection and ordered prefix, the sum of absolute polynomial coefficients is at most B_0. A direct bound from (4.5)–(4.7) counts at most $(m + r)^r$ output occupations,

$(n+1)^r$ input subsets and 2^r suboccupations. Bound $\binom{v}{u} \le 2^r$, the squared-permanent coefficient sum by $(r!)^2$, unused-color counts by $(n+1)^r$, and both color reweightings by $e^{2r} < 9^r$. The projector coefficients contribute C_0^2. Also

$$\prod_i F_{v_i,h_i}(1) = \prod_i \binom{v_i + h_i}{v_i} \le \prod_i (h_i+1)^{v_i} \le (n+1)^r.$$

The inverse lift factor is at most n^{2r}, since $t_{\ell,l,l} \ge \binom{\ell}{l}^{-2}$. Remaining binomial denominators only decrease the bound, and $h!/\ell! \le 1$. The sums over a, j, l and the reference term are absorbed in (5.2). Averaging the T collections does not increase the bound.

If both matrices have entries of modulus at most two, telescoping a degree-$2r$ monomial bounds each prefix change by

$$B_0(2r)2^{2r} \max_{i,j} |U_{ij} - V_{ij}|. \tag{5.3}$$

Round real and imaginary entries to b_{entry} fractional bits. Set

$$\delta = \frac{\varepsilon}{8nm^n}, \qquad 2^{b_{\text{entry}}} \ge \frac{4r2^{2r}B_0}{\delta}. \tag{5.4}$$

The complex entry error is at most $2 \cdot 2^{-b_{\text{entry}}}$, so every prefix error is at most δ, uniformly in U and the colors. The required b_{entry} is polynomial in $n, m, \log(1/\varepsilon)$, uniformly for $r \le n$; Section 8 records the parameter dependence. By Lemma 5.1, the total encoding contribution is at most $2nm^n\delta = \varepsilon/4$, which is below the original $\varepsilon/2$ allowance.

5.3 Exact arithmetic at the encoded input

Evaluate the prefix polynomial at the encoded dyadic matrix with exact Gaussian rational arithmetic. This removes a separate numerical-arithmetic error budget. Factorials and binomials have polynomial bit length. The rational entries of the triangular system (4.1) have $O(r\log(m+r))$ bits. Clearing at most $O(r^2)$ denominators and bounding the resulting determinants gives polynomial bit length for all c_j.

In (4.11), Y-degree is at most r, and matrix-entry degree is at most $2r$. Products of the finite-series factorial denominators have polynomial logarithmic size. A common dyadic input denominator has exponent at most $2rb_{\text{entry}}$. The remaining reduction, coloring, lift and averaging denominators have polynomial bit length; taking their product over the at most polynomially many distinct parameter indices is a sufficient common denominator. Matrix-index summation and the bounded-degree products add only polynomially many magnitude bits. Thus intermediate numerators and denominators have polynomial bit length, despite the $\exp(O(r))$ coefficient count. Exact arithmetic preserves the polynomial running time in (4.12).

5.4 Finite random bits and the final budget

An exact draw from a non-dyadic rational probability does not in general admit a fixed finite number of fair random bits. Replace all such draws explicitly. For a categorical law on $a \le m$ outcomes, round cumulative thresholds down to the grid $2^{-b_{\text{coin}}}$ and use a uniform b_{coin}-bit integer. The TV discrepancy is at most $a2^{-b_{\text{coin}}} \le m2^{-b_{\text{coin}}}$. Threshold comparisons are exact.

One sample uses at most

$$L = T\left(nr + \frac{mr(r+1)}{2}\right) + 2n \tag{5.5}$$

categorical choices: all input and output colors, n Bernoulli loss decisions whose sum is K, and at most n output-mode draws. Choose

$$2^{b_{\mathrm{coin}}} \geq 8mL/\varepsilon. \tag{5.6}$$

A union coupling bounds their total contribution, including the change in the color distribution, by $\varepsilon/8$. Both L and the arithmetic work are polynomial, and $b_{\mathrm{coin}} = O(\log(Lm/\varepsilon))$. There is no unbounded rejection loop. For $m = n = 1$, use the same threshold construction with the two Bernoulli outcomes directly.

Completion of Theorem 1.1. Use (3.11), the finite-ring oracle, (5.4), and the finite-bit choices above. Sample K, use clipped-prefix sampling conditional on it, and return the output occupation. The preceding bounds give

$$\mathbb{E}_U \,\mathrm{TV}(q_U, p_{\eta,U}) \leq \frac{3\varepsilon}{8} + \frac{\varepsilon}{4} + \frac{\varepsilon}{8} = \frac{3\varepsilon}{4} < \varepsilon.$$

Equation (4.12) and the bit bounds prove worst-case polynomial time for fixed η. The input encoding meets the required bit count and error allocation. □

For one sample, compute $r, \tau, T, D(l)$ and the projector coefficients; draw and retain the color collections; sum n finite-bit Bernoulli decisions to obtain K; at each ordered prefix, evaluate its m children using (4.8)–(4.9), clip and normalize, and draw the next mode. Return the occupation of the resulting list. A zero approximate normalization uses any fixed fallback. This summary specifies the mathematical construction; the accompanying small-case checkers are validation tools, not an optimized implementation of this full algorithm.

6 The harmonic spectrum of equal-squeezing inputs

6.1 Photon-number sectors and exact reductions

Put $A^\dagger = \sum_{i=1}^m (a_i^\dagger)^2$. For even N, let $|\Omega_N\rangle$ be the normalized vector proportional to $(A^\dagger)^{N/2}|0\rangle$, and write $\rho_l = R_{N,l}|\Omega_N\rangle\langle\Omega_N|$. Global photon-number dephasing of the squeezed input leaves its eventual photon-count law unchanged: the interferometer conserves number and each loss Kraus operator lowers bra and ket by the same number. Thus the Gaussian experiment is exactly a mixture of these input sectors. Its weights and generating function are

$$\Pr(N = 2p) = (1-\zeta)^{m/2} \frac{(m/2)^{\overline{p}}}{p!} \zeta^p, \qquad \mathbb{E}z^N = \left(\frac{1-\zeta}{1-\zeta z^2}\right)^{m/2}. \tag{6.1}$$

Conditional on N, uniform loss gives $K \sim \mathrm{Bin}(N, \eta)$ and surviving state ρ_K. Uniform loss commutes with passive optics, so this description also applies when loss is placed after the interferometer.

Lemma 6.1 (Pair-state reduction spectrum)**.** *Set $a_{L,h} = 0$ unless $0 \leq h \leq L$ and $L - h$ is even; otherwise set*

$$a_{L,h} = \frac{L!}{2^L((L-h)/2)!\,(m/2)^{\overline{(L+h)/2}}}. \tag{6.2}$$

The eigenvalues of ρ_l are

$$\lambda_{N,l,h} = \frac{a_{l,h}a_{N-l,h}}{a_{N,0}}, \qquad h = l, l-2, \ldots, \tag{6.3}$$

with multiplicities $\binom{m+h-1}{h} - \binom{m+h-3}{h-2}$; the second binomial is zero for $h < 2$.

Proof. Use normalized measure on the real unit sphere. Its even moments are sums of pair contractions divided by $m(m+2)\cdots(m+2q-2)$, as follows by separating the radius and direction of a real standard Gaussian. Applying this formula to a trace-free degree-h harmonic tensor gives eigenvalue (6.2) for the integral kernel $(x\cdot y)^L$ on real spherical harmonics of degree h. The vector $\int x^{\otimes N}\,dx$ is proportional to $|\Omega_N\rangle$ and has squared norm $a_{N,0}$. The embeddings of a spherical harmonic into $\mathrm{Sym}^l\,\mathbb{C}^m$ and $\mathrm{Sym}^{N-l}\,\mathbb{C}^m$ have squared singular values $a_{l,h}$ and $a_{N-l,h}$. Expanding the integral in an orthonormal real harmonic basis gives Schmidt coefficients $\sqrt{a_{l,h}a_{N-l,h}/a_{N,0}}$. This proves the spectrum, including zero eigenvalues and their multiplicities. □

Define $W_l=\Pi_{l,l}\rho_l$ and $b_l=\|W_l\|_{\mathrm{HS}}^2$. Compatibility and Lemma 2.1 imply

$$b_0=1,\qquad b_l=\operatorname{Tr}\rho_l^2-\sum_{h<l}\frac{b_h}{t_{l,h,h}}. \tag{6.4}$$

This gives one exact way to compute the norms. For a bound uniform in the truncation order we next derive their closed form.

6.2 Coherent symbols and an all-order norm formula

For a complex unit vector z, put $x=|z\cdot z|^2$ and $b=(m-1)/2$. Complex spherical monomial moments give $\mathbb{E}x^j=j!/((m+1)/2)^{\overline{j}}$; these moments on $[0,1]$ identify the law as $\mathrm{Beta}(1,b)$. The coherent symbol of the reduced state is

$$\langle z^{\otimes l}|\rho_l|z^{\otimes l}\rangle=\sum_{a=0}^{\lfloor l/2\rfloor}c_{l,a}x^a,\qquad c_{l,a}=\frac{(l!)^2(N/2)_{l-a}((m+N)/2)^{\overline{a}}}{4^a(a!)^2(l-2a)!\,(m/2)^{\overline{l}}(N)_l}. \tag{6.5}$$

One derivation sums over $N=2p$ with unnormalized weight $(m/2)^{\overline{p}}t^p/p!$. Differentiating the quadratic Gaussian exponential gives the normally ordered moment

$$\langle(a_z^\dagger)^l a_z^l\rangle=\sum_a\frac{(l!)^2x^a}{4^a(a!)^2(l-2a)!}\left(\frac{t}{1-t}\right)^{l-a}(1-t)^{-a}. \tag{6.6}$$

Multiply by $(1-t)^{-m/2}$, extract t^p, and divide by the sector weight and $(N)_l$. Cancelling the rising factorials yields (6.5). This uses the usual Gaussian moment calculus [21] without a large-mode approximation.

Let P_j be the monic degree-j polynomial orthogonal for $\mathrm{Beta}(1,b)$. For $l=2j$, the top-sector symbol is $c_{l,j}P_j(x)$. Indeed, the $O(m)$-invariant coherent symbols are polynomials in x; the invariant components of lower $U(m)$ sectors span the lower-degree polynomials. Inequivalent sectors are orthogonal for the coherent-symbol inner product by equivariance. At odd l, the available polynomial degree has not increased, so the invariant top component is zero. The monic shifted Jacobi norm is

$$H_j=\int_0^1 P_j(x)^2b(1-x)^{b-1}\,dx=\frac{b}{2j+b}\binom{2j+b-1}{j}^{-2}. \tag{6.7}$$

It follows from the Rodrigues formula and j integrations by parts. The coherent-frame eigenvalue on the top $U(m)$ sector is

$$C_l=\frac{(l!)^2}{m^{\overline{2l}}}. \tag{6.8}$$

To see its normalization, evaluate the coherent-symbol norm of the unit Hilbert–Schmidt highest-weight matrix $|l_1\rangle\langle l_m|$: the spherical integral is $\mathbb{E}|z_1|^{2l}|z_m|^{2l} = C_l$. Schur orthogonality gives the same scalar on the full irreducible sector. Hence

$$b_{2j} = c_{2j,j}^2 H_j / C_{2j}, \qquad b_{2j+1} = 0. \tag{6.9}$$

Dividing consecutive even expressions gives, for even $0 \le l \le N-2$,

$$\frac{b_{l+2}}{b_l} = \frac{(l+1)^2(m+l-1)^2(N+m+l)^2}{(N-l-1)^2(m+2l-1)(m+2l)(m+2l+1)(m+2l+2)}. \tag{6.10}$$

Thus the norm formula holds at every order; the finite checks in Section 11 are separate controls.

6.3 Geometric decay on a typical loss sector

For $0 \le l \le k \le N$, define the exact Haar error weights

$$A_l(m, N, k) = \frac{D_k f_l b_l}{t_{k,l,l}}. \tag{6.11}$$

They sum to the D_k-weighted mean squared error when the corresponding sectors are omitted. Here $A_0 = 1$ and odd weights vanish. Substituting (2.5) into (6.10) yields

$$\frac{A_{l+2}}{A_l} = \frac{l+1}{l+2}\frac{m+l-1}{m+l}\frac{(k-l)(k-l-1)}{(N-l-1)^2}\frac{(N+m+l)^2}{(m+k+l)(m+k+l+1)}. \tag{6.12}$$

Suppose $k \le N-1$. The functions $(k-l)/(N-l-1)$ and $(N+m+l)/(m+k+l)$ are nonincreasing in l. Each ratio in (6.12) is therefore at most $\widetilde{\rho}^2$, where

$$\widetilde{\rho} = \frac{k}{N-1}\frac{N+m}{m+k}. \tag{6.13}$$

Whenever $\widetilde{\rho} < 1$, $A_{2j} \le \widetilde{\rho}^{2j}$. Let $p_{N,k,U}$ be the exact k-photon law and $q_{N,k,r,U}$ its normalized signed harmonic truncation. Cauchy–Schwarz and the exact measurement frame give

$$\mathbb{E}_U\|p_{N,k,U} - q_{N,k,r,U}\|_1 \le \left(\sum_{\substack{l>r \\ l \ meven}} A_l\right)^{1/2} \le \frac{\widetilde{\rho}^{(r+1)/2}}{\sqrt{1-\widetilde{\rho}^2}}. \tag{6.14}$$

The exponent is conservative and valid for either parity of r.

For constants $B > 0$ and $\eta' < 1$, assume $N/m \le B$ and $k/N \le \eta'$. Set

$$\rho = \frac{\eta'(1+B)}{1+\eta' B}, \qquad \rho_* = (1+\rho)/2 < 1, \qquad N_0 = \left\lceil \frac{2\rho_*}{1-\rho} \right\rceil. \tag{6.15}$$

If $N \ge N_0$, then $\widetilde{\rho} \le (N/(N-1))\rho \le \rho_*$. Consequently the cutoff needed in (6.14) is $O_{B,\eta'}(1 + \log(1/\varepsilon))$, independent of the mode and photon counts. The next section handles the complementary sectors with their actual probabilities.

7 A bounded Gaussian sampler and its hafnian oracle

7.1 Finite photon draws and exceptional sectors

Fix ζ, η as in Theorem 1.2 and put $\delta = \varepsilon/64$. Choose a fixed rational $z > 1$ with $\zeta z^2 < 1$, for instance $z = 1 + (1-\zeta)/4$. Set $H = ((1-\zeta)/(1-\zeta z^2))^{1/2}$ and choose a fixed integer B with $Hz^{-B} \le 1/2$. Markov's inequality applied to (6.1) gives

$$\Pr(N > Bm + L) \le 2^{-m} z^{-L}. \tag{7.1}$$

Take $L = \lceil \log(1/\delta)/\log z \rceil$ and $N_{\max} = \lceil Bm + L \rceil$. Draw from the finite list of even $N \le N_{\max}$ plus one tail category, returning vacuum on the latter. This alters the target law by at most δ. The finite probabilities have a rational recurrence and one common factor $(1-\zeta)^{m/2}$, rational for even m and quadratic algebraic for odd m.

Use $\eta' = (1+\eta)/2$ in (6.15). Choose an integer $r \ge 1$ such that

$$\rho_*^r \le \delta^2(1-\rho_*^2), \qquad m_{\mathrm{small}} = \max\{r, \lceil \log_2(1/\delta) \rceil\}, \tag{7.2}$$

and set

$$N_{\mathrm{small}} = \max\left\{ r, N_0, \left\lceil \frac{\log(1/\delta)}{2(\eta'-\eta)^2} \right\rceil \right\}. \tag{7.3}$$

If $m < m_{\mathrm{small}}$, then $N_{\max} = O_{\zeta,\eta}(1 + \log(1/\varepsilon))$; use the small-photon method below for every retained N. Otherwise return vacuum when $N > Bm$, an event of probability at most $2^{-m} \le \delta$. Draw $K \sim \mathrm{Bin}(N, \eta)$. Use the small-photon method when $N \le N_{\mathrm{small}}$. In the remaining branch, return vacuum when $K > \eta' N$, whose conditional probability is at most δ by Hoeffding's inequality. The retained large branch has $m \ge r$, $N/m \le B$, $N \ge N_0$ and $k/N \le \eta'$. The three exceptional events cost at most 3δ in TV. All branches have a predetermined finite number of steps.

7.2 Pair operators and the small-photon method

Write the monic polynomial from Section 6 as $P_j(x) = \sum_a p_{j,a} x^a$. For even $l = 2j$, put

$$d_{l,a} = \frac{c_{l,j} p_{j,a}}{(l)_{2a}}, \qquad W_l = \sum_{a=0}^{j} d_{l,a} (A^\dagger)^a A^a. \tag{7.4}$$

The operator identity follows because both sides have coherent symbol $c_{l,j} P_j(|z \cdot z|^2)$, and coherent symbols determine operators on $\mathrm{Sym}^l \mathbb{C}^m$. After the interferometer, replace $A^\dagger$ by $A_B^\dagger = \sum_{ij} B_{ij} a_i^\dagger a_j^\dagger$, with $B = UU^T$. For an occupation v of size l, its diagonal is

$$Z_l(v) = \sum_a d_{l,a} 4^a (a!)^2 \sum_{\substack{u \le v \\ |u| = 2a}} \binom{v}{u} \frac{|\operatorname{Haf}(B_u)|^2}{u!}. \tag{7.5}$$

Here B_u repeats row and column i exactly u_i times. Expansion of $(x^T B x)^a$ gives the factor $4^a (a!)^2$.

For the unprojected reduced state ρ_j, the same formula uses $d_{j,a}^{\mathrm{full}} = c_{j,a}/(j)_{2a}$. At a particular ordered prefix of occupation h and length j, multiply its Fock diagonal by $h!/j!$. There are at most $\prod_i (h_i + 1) \le 2^j$ subvectors $u \le h$. A subset dynamic program evaluates each hafnian on at most j labels in $O(j^2 2^j)$ operations, pairing one fixed remaining label with each other label. At most mN prefix queries therefore cost $\mathrm{poly}(m, N) 4^N$. In the small branch $N = O_{\zeta,\eta}(1 + \log(1/\varepsilon))$, so this cost is polynomial in the desired variables and includes all collisions.

7.3 A collision-resolving low-degree oracle

For the large branch, use the output masks $a_l(v)$ of (3.5), with the same deficit cutoff and singleton colorings. No input coloring is needed because (7.4) supplies W_l exactly. With one complete set of colors fixed, define

$$\mathcal{A}_l(U) = \Pi_{l,l}\mathcal{D}_{a_l}(\Gamma_l(U)W_l\Gamma_l(U)^*), \qquad B_j(U) = \frac{I}{D_j} + \sum_{\substack{2\le l\le \min(r,j)\\ l\ meven}} J_{j,l}\mathcal{A}_l(U). \tag{7.6}$$

This family is diagonal, Hermitian, trace one and particle compatible. Average T independent complete color collections, retaining them for every prefix query of a sample.

Denote the rational top-projector coefficients of (4.1) by $\gamma_b^{(l)}$ to distinguish them from (6.5). The outer kernel for a prefix occupation h, $|h| = \ell$, is

$$K_{\ell,l}(h,v) = \sum_{b=0}^{l} \frac{\gamma_b^{(l)}}{\binom{l}{b}\binom{\ell}{b}}[x^b]\prod_i \sum_t \binom{v_i}{t}\binom{h_i}{t}x^t. \tag{7.7}$$

Equation (4.6) then expresses the ordered degree-l prefix as

$$\frac{h!}{\ell!\, t_{\ell,l,l}} \sum_{|v|=l} a_l(v)Z_l(v)K_{\ell,l}(h,v). \tag{7.8}$$

We evaluate this sum without listing all D_l occupations.

Fix a retained heavy support H and occupancies $b_i \ge 2$, and let $s = l - \sum_{i\in H} b_i$. Use commuting square-zero variables $X_1, \dots, X_s$, variables z_i capped at degree b_i, and variables x, y capped at degrees $l, \lfloor l/2\rfloor$. For the singleton coloring $g : [m] \to [s]$, set

$$Z_i = X_{g(i)}(1 + h_i x) \quad (i \notin H), \qquad Z_i = z_i \quad (i \in H),$$

and write $Z = \mathrm{diag}(Z_i)$. If $s = 0$, set $Z_i = 0$ off H. Form

$$G = \prod_{i\notin H}(1 + X_{g(i)}(1 + h_i x))\, \det(I - yZBZ\overline{B})^{-1/2} \times \prod_{i\in H}\sum_t \binom{b_i}{t}\binom{h_i}{t}x^t. \tag{7.9}$$

The first product is one when $s = 0$. The formal identity behind this expression is

$$\det(I - yZBZ\overline{B})^{-1/2} = \sum_{u:\,|u|\ meven} \frac{|\operatorname{Haf}(B_u)|^2}{u!}y^{|u|/2}Z^u. \tag{7.10}$$

The square root is the formal branch with constant term one. To prove it, take the Gaussian Bargmann inner product of the quadratic exponential $\exp(tz^TBz/2)$ with itself. Expanding the exponentials yields the hafnian sum; evaluating the Gaussian integral yields the determinant. Equality near zero implies equality of formal coefficients. This established Gaussian identity [19, 21] needs symmetry of B, not unitarity.

Extract coefficients of $X_1 \cdots X_s \prod_{i\in H} z_i^{u_i} y^a x^b$ from (7.9), multiply by

$$d_{l,a}4^a(a!)^2 \prod_{i\in H}\binom{b_i}{u_i}, \frac{\gamma_b^{(l)}}{\binom{l}{b}\binom{\ell}{b}p_s}, \qquad p_s = s!/s^s, quad p_0 = 1, \tag{7.11}$$

and sum over all allowed indices and heavy patterns. Multiplying by $h!/(\ell! t_{\ell,l,l})$ gives (7.8). Square-zero variables allocate distinct singleton colors between the active hafnian and its

spectators. Heavy-mode factorials remain in (7.10), and the explicit binomial weights restore their choices in (7.5). The coefficient of x^b gives the outer kernel. These observations prove the identity for every l.

The ring has dimension at most

$$(l+1)(\lfloor l/2\rfloor+1)2^s\prod_{i\in H}(b_i+1)\le \mathrm{poly}(l)2^l. \tag{7.12}$$

Evaluate the determinant factor as $\exp(\sum_{a\ge1}\mathrm{Tr}(M^a)/(2a))$ with $M=yZBZ\overline{B}$; its positive y-degree makes the series finite. Matrix operations in this ring cost $\mathrm{poly}(m,l)\exp(O(l))$. Lemma 3.1 bounds heavy-support enumeration by $\tau^{-1}\exp(O(l))$ times a polynomial. Including T collections and the prefix path gives total arithmetic cost

$$\mathrm{poly}(m,N,r)T\tau^{-1}\exp(O(r)). \tag{7.13}$$

7.4 Bias, variance and proper sampling

Schur orthogonality gives, at each l-photon label v,

$$\mathbb{E}_U|Z_l(v)|^2=\frac{b_l}{d_l}\|\Pi_{l,l}|v\rangle\langle v|\|_{\mathrm{HS}}^2\le b_l/d_l. \tag{7.14}$$

At most τD_l labels are omitted; for $m\ge l$ we have $D_l/d_l\le 3f_l$. Applying the outer projection and lift, the D_k-weighted squared bias is therefore at most $3\tau A_l(m,N,k)$. The inverse singleton inclusion probability is at most e^l. Since the squared diagonal norm has Haar mean f_lb_l, the corresponding variance of T independent collections is at most $T^{-1}e^lA_l$. Different harmonic degrees remain orthogonal after dephasing. Combining these bounds with (6.14) gives

$$\mathbb{E}_{U,\mathcal{C}}\|p_{N,k,U}-q_{N,k,U,\mathcal{C}}\|_1\le\frac{\rho_*^{(r+1)/2}}{\sqrt{1-\rho_*^2}}+\sqrt{\frac{3\tau}{1-\rho_*^2}}+\sqrt{\frac{e^r}{T(1-\rho_*^2)}}. \tag{7.15}$$

Choose r as in (7.2), $\tau\le\delta^2(1-\rho_*^2)/3$ and $T\ge 3^r/[\delta^2(1-\rho_*^2)]$. Each term is at most δ. All comparisons can use rational bounds to the fixed constants. Lemma 5.1 turns the compatible ordered law into a proper sampler with mean TV error at most $3\delta/2$. Mixing over its retained color collections preserves this bound by convexity.

7.5 Matrix precision for both branches

Rounding U need not preserve consistency of the unprojected small-photon formulas. We apply the perturbation part of Lemma 5.1 directly to both branches. If every queried ordered child mass has absolute error at most α, the terminal TV change is at most $2km^k\alpha$. The exponential factor is used only to choose precision; the algorithm visits at most km children.

Suppose $\max_{ij}|V_{ij}-U_{ij}|\le\beta\le1$. Then $|V_{ij}|\le2$, $|(VV^T-UU^T)_{ij}|\le3m\beta$, and entries of both pair matrices are bounded by $4m$. A hafnian on $2a$ labels has $(2a)!/(2^aa!)$ monomials of degree a. Telescoping products in (7.5), and using $\sum_{u\le h}\binom{h}{u}=2^j$, bounds the change of any j-photon diagonal by $L_j\beta$, where

$$L_j=6j2^j\sum_{a=1}^{\lfloor j/2\rfloor}|d_{j,a}^{\mathrm{full}}|((2a)!)^2(4m)^{2a},\qquad L_{\mathrm{small}}=\max\{1,L_0,\dots,L_N\}. \tag{7.16}$$

Here $L_0=L_1=0$. Multiplication by $h!/j!\le1$ preserves the bound. Therefore

$$\beta\le\frac{\delta}{4Nm^NL_{\mathrm{small}}} \tag{7.17}$$

gives at most $\delta/2$ matrix-encoding error for every $k \le N$ in the small branch. All these rational constants have polynomial bit length in N and $\log(m+N+2)$.

For the large branch, define $L_Z(l)$ by (7.16) with $d_{l,a}$ from (7.4) in place of $d^{\mathrm{full}}_{j,a}$. Set

$$C'_l = \sum_{b=0}^{l} |\gamma_b^{(l)}|, \qquad T'_l = \max_{l \le j \le N} t^{-1}_{j,l,l},$$
$$L_{\mathrm{large}} = 1 + \sum_{\substack{2 \le l \le \min(r,N) \\ l \ meven}} D_l 3^l 2^l (N+1)^l C'_l T'_l L_Z(l). \tag{7.18}$$

Each mask weight is at most 3^l, the kernel in (7.7) is bounded by $2^l(N+1)^l C'_l$, and there are D_l occupations. The lift contributes T'_l. Thus (7.18) bounds the derivative of every ordered prefix, uniformly over colors; averaging collections does not enlarge it. These constants require polynomially many factorial expressions, not enumeration of D_l labels. Their logarithms are polynomial in $N, r, \log(m+N+2)$. Taking

$$\beta \le \frac{\delta}{4Nm^N L_{\mathrm{large}}} \tag{7.19}$$

again gives at most $\delta/2$ encoding error. Before drawing N, choose a single entry precision sufficient for all even $N \le N_{\max}$ and every applicable branch. This requires polynomially many comparisons and $\mathrm{poly}_{\zeta,\eta}(m, \log(1/\varepsilon))$ bits. Vacuum requires no optical queries.

After matrix rounding, evaluate the formulas in exact Gaussian rational arithmetic. The beta moments are rational; clearing the denominators of their moment systems and bounding determinants gives polynomial bit length for the monic polynomial coefficients. The factorial, projector and lift coefficients have the same property. The finite ring has matrix degree at most $2r$, while the small branch has degree at most $2N$. A common dyadic denominator therefore has polynomial exponent. The remaining denominator families have polynomially many parameter indices; their product still has polynomial logarithmic size. Bounded-degree products and mode-index sums add polynomially many magnitude bits. Thus exact intermediate words have polynomial length in both branches.

7.6 Finite random bits and completion of the Gaussian theorem

For a rational categorical law on d outcomes, inverse-CDF thresholds rounded to a b-bit grid incur TV error at most $d2^{-b}$. Let M_{draw} be a polynomial upper bound on the number of categorical draws: the initial sector choice, at most $N_{\max}$ loss draws, $O(Tmr^3)$ color assignments and at most $N_{\max}$ output choices. With $d_{\max}$ bounding their alphabet sizes, take

$$b \ge \lceil \log_2(2M_{\mathrm{draw}} d_{\max}/\delta) \rceil . \tag{7.20}$$

A coupling bounds the total finite-bit contribution by $\delta/2$. Computing the number weights in (6.1) to interval accuracy $\delta/(4(N_{\max}+1))$ contributes at most another δ; their one square root is computable with polynomially many bits. All thresholds use bounded computations and a fixed number of random bits. For $m = 1$, the number-sector and binomial draws already give the whole output law.

Completion of Theorem 1.2. Use the finite number draw and branch rules above. In the small branch, use the full pair-state prefix formulas. In the typical large branch, evaluate (7.6) by the finite-ring oracle and use clipped-prefix sampling with fixed colors. The total mean TV error is bounded by

$$3\delta + \frac{3}{2}\delta + \frac{1}{2}\delta + \frac{1}{2}\delta + \delta = \frac{13}{2}\delta < \varepsilon,$$

accounting respectively for exceptional sectors, harmonic and color error, matrix encoding, bounded random draws and number-weight arithmetic. Here $N_{\max} = O_\zeta(m + \log(1/\varepsilon))$, $N = O_{\zeta,\eta}(1 + \log(1/\varepsilon))$ on the small branch, and $r = O_{\zeta,\eta}(1 + \log(1/\varepsilon))$ on the large branch. Also τ^{-1} and T are polynomial in $1/\varepsilon$. The small cost $\mathrm{poly}(m, N)4^N$, the large cost (7.13), and the preceding word-length bounds prove the worst-case polynomial bit claim. □

The proof uses equal squeezing in all modes in the pair-state spectrum and in the bound on N/m. Adding vacuum inputs changes the ambient harmonic decomposition; unequal squeezing changes the pair-state symmetry. The theorem does not extend to either setting by this argument. Its cost constants also depend on both ζ and η, and no uniform Gaussian loss–accuracy phase boundary is claimed here.

8 Loss and accuracy as input parameters

The construction has a uniform bound when the transmission is supplied as a rational input. This makes the range of decreasing losses precise. Put $s = 1 - \eta$, $A = 288e^4$, and let L_η, L_ε be the binary lengths of the rational parameters.

Proposition 8.1 (Uniform parameter bound)**.** *There is a single polynomial P, independent of η and ε, such that the sampler in Sections 4–5 uses at most*

$$P(n, m, L_\eta, L_\varepsilon)\, \varepsilon^{-4} s^{-1} A^r \tag{8.1}$$

bit operations and has $\mathbb{E}_U \mathrm{TV}(q_U, p_{\eta,U}) \le \varepsilon$, where

$$r = \min\left\{ n, \left\lceil \frac{\log(64/(\varepsilon^2 s))}{-\log \eta} \right\rceil \right\}. \tag{8.2}$$

The statement holds for every rational $0 < \eta, \varepsilon < 1$ and every $m \ge n \ge 1$. The input matrix precision and each intermediate arithmetic word length are bounded by a universal polynomial in $n, m, L_\eta, L_\varepsilon$.

Proof. All error estimates preceding the choice (3.11) hold pointwise in η. They use $\sum_{l \ge 1} \eta^l \le s^{-1}$, rather than a constant hidden in a fixed-transmission limit. With $\tau = \varepsilon^2 s/192$ and $T = \lceil 64\varepsilon^{-2} r 9^r \rceil$, we have $T \le 65\varepsilon^{-2} r 9^r$. Substituting these bounds in (4.13) gives

$$\mathrm{poly}(n, m, r)\, \varepsilon^{-4} s^{-1} (288e^4)^r$$

arithmetic operations, with a universal polynomial factor. The case $r = n$ has zero harmonic tail and satisfies the same error allocation.

We check that arithmetic has not hidden any nonuniform dependence. The quantities in (5.2) obey

$$\log B_0 = O\Big(r^2 \log(m + r + 2) + r \log(n + 2)\Big).$$

Consequently (5.4) permits

$$b_{\mathrm{entry}} = O\Big(n \log(m + 1) + r^2 \log(n + m + r + 2) + \log(1/\varepsilon)\Big) \tag{8.3}$$

fractional bits, with an absolute implied constant. Since $r \le n$, this is a universal polynomial bound. The rational triangular system has only polynomially many entries of $O(r \log(m + r))$ bits. Clearing its denominators and applying the determinant formula bounds the bit lengths of the projector coefficients uniformly. The finite ring has total degree at most r; its factorial denominators and the dyadic input denominator therefore require polynomially many bits.

The remaining rational factors are indexed by the polynomially many degree and prefix-length parameters. Their product is a valid common denominator of polynomial bit length. Finally, $\log T = O(r + \log(r+1) + \log(1/\varepsilon))$, so averaging does not change this conclusion.

No logarithm oracle is needed to choose (8.2): compare successive rational powers η^j with $\varepsilon^2 s/64$, stopping at $j = n$. These at most n comparisons use $O(nL_\eta + L_\varepsilon)$-bit integers. The collision cutoff is likewise chosen by rational binomial comparisons. The required precision for each categorical draw is $O(\log(Lm/\varepsilon))$ with L as in (5.5); hence it is polynomial in the same inputs. The Bernoulli thresholds for loss are rational input data. Polynomial-time integer arithmetic applied to the preceding operation count proves (8.1). The final error budget in Section 5 is unchanged. □

Proposition 8.2 (A regime with vanishing loss)**.** *Fix $c > 0$ and $b \geq 0$. Suppose, for sufficiently large n, that*

$$1 - \eta \geq c\,\frac{\log\log n}{\log n}, \qquad \varepsilon^{-1} = O((\log n)^b). \tag{8.4}$$

Then the full-output sampler has worst-case running time polynomial in $n, m, L_\eta, L_\varepsilon$, for every $m \geq n$. In particular this includes every fixed positive accuracy tolerance. More generally, writing $N = n + m + \varepsilon^{-1} + 2$, a sufficient condition is

$$\frac{\log(64/(\varepsilon^2(1-\eta)))}{-\log\eta} = O(\log N), \qquad (1-\eta)^{-1} = N^{O(1)}. \tag{8.5}$$

Proof. Use $-\log\eta \geq s$. Under (8.4),

$$r \leq \left(\frac{2b+1}{c} + o(1)\right)\log n, \qquad s^{-1} = O(\log n/\log\log n).$$

Thus A^r is polynomial in n, and the other factors in (8.1) are polynomial as well. Condition (8.5) gives the same conclusion with N in place of n. □

The accuracy assumption matters. If $s = c\log\log n/\log n$ and $\varepsilon = n^{-a}$ for fixed $a > 0$, the prescribed degree is

$$r = \left(\frac{2a}{c} + o(1)\right)\frac{(\log n)^2}{\log\log n}.$$

The prescribed number of color collections is then superpolynomial. This describes the present algorithm, not a hardness threshold.

Parameter-dependent truncation is already explicit in Go–Jeong [7], Theorem 2 and equation (19), and their Theorem 3 treats transmission loss. Proposition 8.1 carries that type of parameter dependence through a construction valid at arbitrary mode density and with all collisions retained; the variable-loss substitution alone is not a separate novelty claim. These upper bounds also do not close a loss phase diagram. The fixed-number-loss hardness result of Aaronson–Brod [15] uses an inverse-polynomial accuracy requirement and additional complexity assumptions. The linear-mode ideal-boson-sampling hardness theorem of Bouland et al. [16] has its own conjectural and accuracy hypotheses. Neither supplies a matching lower bound for (8.4).

9 A restricted partial-distinguishability extension

Consider photons with internal states

$$|\phi_i\rangle = \sqrt{\xi_i}|0\rangle + \sqrt{1-\xi_i}|i\rangle, \qquad 0 \leq \xi_i \leq 1, \tag{9.1}$$

where the $n+1$ internal vectors are orthonormal. The interferometer and losses act identically on these internal modes, and detectors count photons without resolving the internal state. This is the heterogeneous orthogonal bad-bit model studied in [17]; its overlap matrix has $S_{ii} = 1$ and $S_{ij} = \sqrt{\xi_i \xi_j}$ for $i \neq j$.

Proposition 9.1. *For rational $\xi_i \in [0,1]$, fixed rational $0 < \eta < 1$, and every $m \geq n$, the complete output distribution of (9.1) with independent transmission η admits a sampler with Haar-average TV error at most ε. Its worst-case running time is polynomial in $n, m, 1/\varepsilon$ and the binary input length. For variable η, it makes at most one call to the sampler of Proposition 8.1, with accuracy $\varepsilon/2$, plus polynomial work.*

Proof. Expand the input by the subset C of photons in the common internal mode. Distinct C have different occupations of at least one private internal mode. These sectors remain orthogonal after passive optics; including the loss environment preserves the distinction before the internal and environmental modes are traced out. Thus the spatial count distribution is exactly a classical mixture with

$$\Pr(C) = \prod_{i \in C} \xi_i \prod_{i \notin C} (1 - \xi_i).$$

Conditional on C, its photons are indistinguishable, whereas every photon outside C has its own orthogonal internal label. Their spatial distributions are independent conditional on U and C, and their occupations are added.

Sample C by independent Bernoulli draws. Simulate the photons in C using Theorem 1.1; each remaining photon is lost with probability $1 - \eta$, and otherwise chooses mode a with probability $|U_{ai}|^2$. An empty common sector returns vacuum. For any fixed C, a permutation of the input columns makes it the first $|C|$ columns. Haar measure is invariant under this permutation. The theorem therefore bounds its average TV error by $\varepsilon/2$, uniformly in C. Taking a product with the private-photon law, adding occupations, and averaging over C cannot increase this error. Sharing the same U creates no independence assumption between different interferometers.

The ξ_i, loss and private-output draws can be implemented with finite rational thresholds. Choose their joint rounding error at most $\varepsilon/4$. A rounded matrix entry error h changes a normalized private-output column law by at most $3mh$ for sufficiently small h. Requiring additionally $h \leq \varepsilon/(12nm)$ bounds all private-output encoding errors by $\varepsilon/4$ and needs only logarithmically many extra bits. The common-sector call already includes its encoding and random-bit budget. These bounds give the stated total error and cost. □

This is a closure consequence of the indistinguishable-photon theorem. The internal-state model and its classical-sector expansion are inherited. The comparison in [17], Section II, uses noncollisional outputs and an independent-Gaussian matrix approximation; Proposition 9.1 retains full collision outcomes at every $m \geq n$ under constant loss. It does not exploit distinguishability to improve the loss exponent, and does not cover arbitrary overlap matrices: their cycle phases need not admit the positive subset mixture above. For example, the three internal states $|0\rangle$, $(|0\rangle + |1\rangle)/\sqrt{2}$, and $(|0\rangle + i|1\rangle)/\sqrt{2}$ have the same pairwise overlap magnitudes as a uniform bad-bit model but a different three-cycle phase. Their three-mode Fourier output distributions differ.

The Gaussian theorem requires a separate input estimate. For example, the two-photon sector of m equally squeezed modes is $m^{-1/2} \sum_i |2_i\rangle$, with one-particle reduction I/m and top harmonic norm $1 - 1/D_2$. For $m \geq 3$ this exceeds the collision-free input estimate $\binom{m}{2}^{-1}$. Sections 6–7 replace that estimate by the pair-state norm formula and its geometric tail; the distinguishability mixture above is not used there.

10 Full-output Haar experiments

10.1 Single-photon inputs

We tested six geometries,

$$(m, n) \in \{(2, 2), (3, 3), (4, 3), (6, 3), (4, 4), (5, 4)\},$$

using 256 independent Haar matrices per geometry (1,536 in total), seed 20261101, and $\eta \in \{0.25, 0.5, 0.75, 0.9\}$. Every survivor number, including vacuum, and every Fock occupation, including collisions, was enumerated. The bosonic representation matrices were built by creation-operator recursion. Harmonic projectors were obtained from QR decompositions of the row spaces of particle reduction, and compatible lifts by linear solves. These constructions avoid inserting the closed harmonic eigenvalue formula into the measured left-hand side.

For the ideal truncated law we varied $r = 0, \ldots, n$. For the randomized signed law we used independent uniform input and output colors, $r = 1, \ldots, n$, and $T = 1, 4, 16, 64, 256$. A complete color collection was retained across all prefix queries. We computed the induced proper sampling law by propagating probabilities through the clipped-prefix recursion, so TV was evaluated without finite-output-shot error. We tested both the full harmonic collision mask and a diagnostic mask omitting harmonic occupations with repeated modes. The latter still retains physical output collisions; it deliberately introduces additional harmonic bias.

The randomized statistic is

$$\mathbb{E}_{U,\mathcal{C}} \operatorname{TV}(S_{U,\mathcal{C}}, p_U), \tag{10.1}$$

where $\mathcal{C}$ denotes the colors. By convexity it upper-bounds the Haar-average TV of the algorithm whose output law also averages over colors. The parameters r, T were varied independently; these small-system runs do not execute the much larger worst-case accuracy prescription.

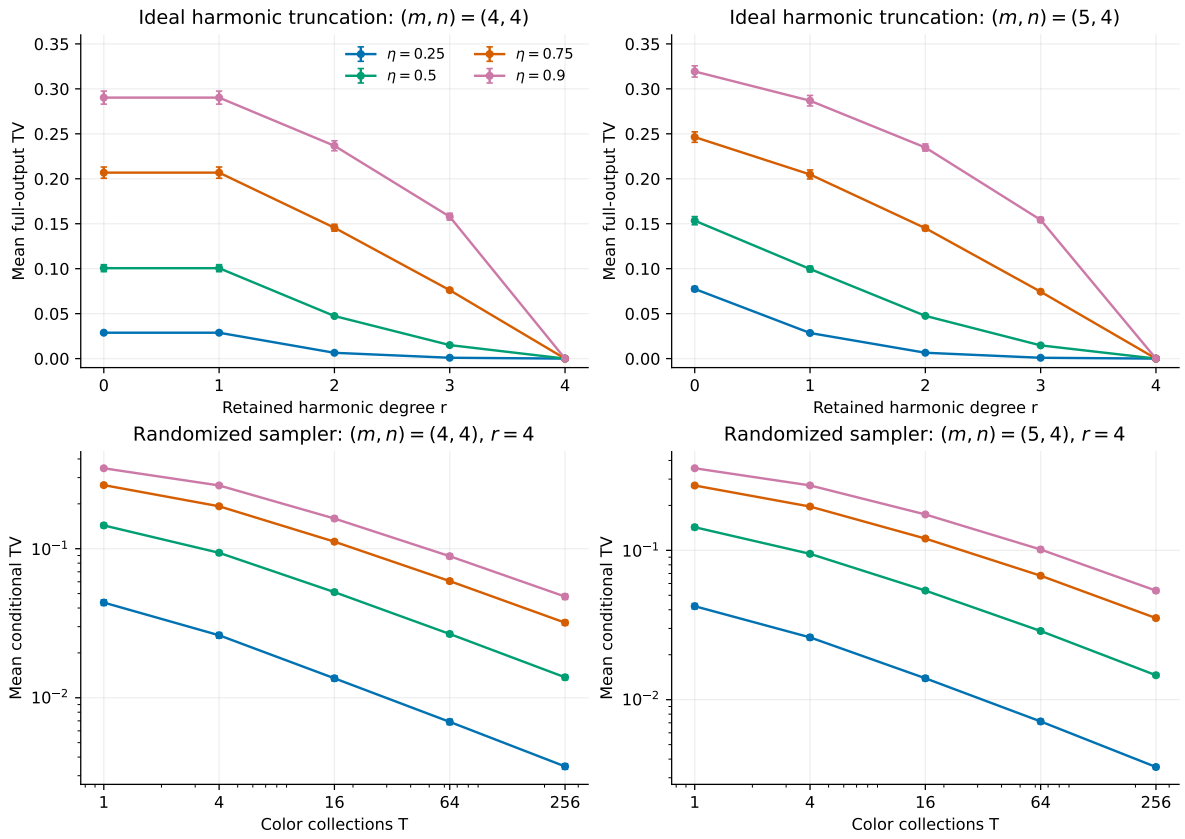


Figure 1: Full-output TV for four input photons. Top: the ideal truncated signed law followed by clipped-prefix sampling. Bottom: the randomized sampler with all harmonic collision patterns retained. Points average 256 independent Haar-and-color experiments; bars show two pointwise standard errors. Different r, T, η values reuse the same interferometers, so their errors are correlated.

At $(m, n) = (4, 4)$ and $\eta = 0.5$, ideal truncation at $r = 2, 3$ gave mean TV 0.04736 and 0.01504. With $r = 4$ and all harmonic collision patterns retained, increasing T from 1 to 256 reduced the measured mean from 0.14330 to 0.01372. At $\eta = 0.9$, the corresponding randomized means were 0.34642 and 0.04773. These observations show convergence of the tested finite construction; they do not estimate its asymptotic running-time exponent. The analytic bound is substantially looser than these means and often exceeds one for small T.

We also compared the measured weighted squared truncation error with its exact Haar expectation,

$$\sum_{k=0}^{n} \binom{n}{k} \eta^k (1-\eta)^{n-k} D_k \sum_{l=r+1}^{k} \frac{\|P_{k,l}\rho_{k,n}\|_2^2}{D_l + D_{l-1}}, \tag{10.2}$$

where $P_{k,l}$ is the degree-l harmonic projector. Among the 76 nonzero parameter cells, the largest absolute difference between measured and predicted means was 1.870 Monte Carlo standard errors. These are correlated pointwise comparisons, not 76 independent confirmations or a simultaneous confidence statement.

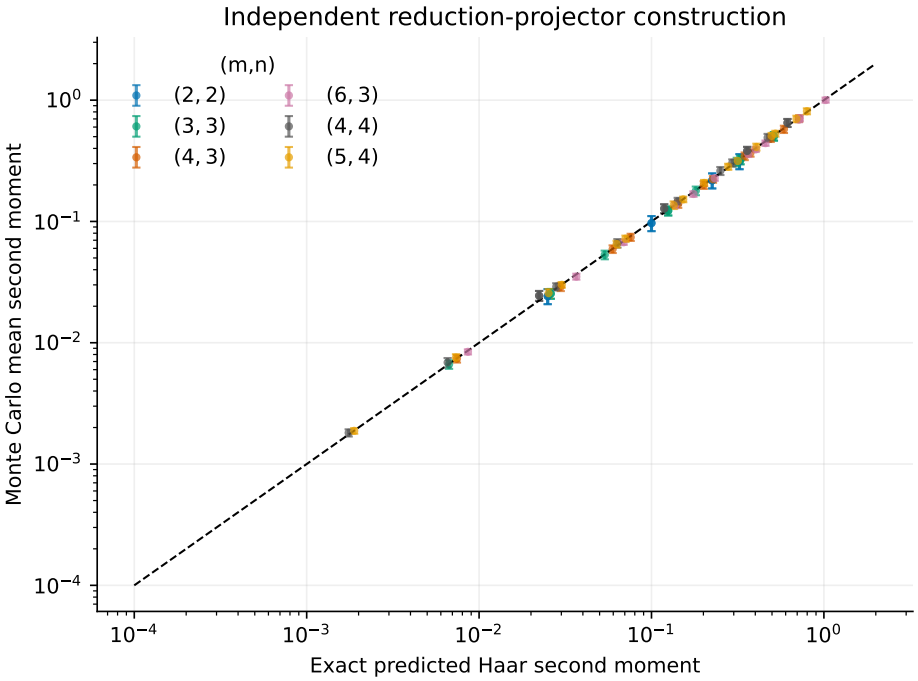


Figure 2: Exact Haar second-moment prediction versus the independently constructed numerical left-hand side. Bars show two pointwise standard errors. Several points overlap because the same sector can vanish in more than one parameter cell. The dashed line is equality.

The largest bosonic-unitarity residual was 5.11×10^{-15} and the largest prefix-consistency residual was 1.78×10^{-15}. Direct permanent controls and the earlier coefficient oracle agreed with the new constructions to below 8×10^{-16}. The sampling inequality (5.1) held pointwise up to 5.0×10^{-16}. Removing the outer harmonic projection in a diagnostic example produced a reduction residual 0.05974, while the correct expression had residual below 2×10^{-17}.

A separate 36-case calculation compared the double-permutation formula for partially distinguishable photons with the subset-mixture construction in Proposition 9.1, including loss and all output collisions. The maximum L_1 difference was 9.23×10^{-16}. A three-cycle-phase control with equal pairwise visibilities gave TV 0.09157 at transmission 0.8, detecting the invalid extension to a general overlap matrix.

All these experiments use double-precision arithmetic and an exponential full-Fock enumeration backend. They are finite implementation and normalization checks, not interval certificates or an implementation of the efficient all-size oracle. The original rational algebraic checks remain separate. Raw observations, seeds, formulas, plotting code and replay instructions accompany this release.

10.2 Conditional Gaussian pair inputs

A second experiment used 128 independent Haar matrices for each of four geometries,

$$(m, N) \in \{(2, 6), (3, 6), (4, 6), (3, 8)\},$$

for a total of 512, with seed 2026110301. It starts from the normalized pair state $|\Omega_N\rangle$ and mixes all survivor sectors at $\eta \in \{0.5, 0.8, 0.95\}$. Every output collision is included. For even cutoffs $0 \le r \le N$, we constructed the full bosonic action and reduction projectors numerically, then computed the exact finite proper law induced by clipped prefixes. This tests the ideal Gaussian harmonic truncation; it does not test the large color counts prescribed by the worst-case algorithm or the unbounded squeezed-state number mixture.

Across 6,528 correlated parameter observations, the nonzero squared-error means differed from the exact weights (6.11) by at most 1.385 estimated standard errors. The maximum prefix residual was 8.89×10^{-16} and the maximum numerical violation of (5.1) was 2.45×10^{-15}. At $(m, N) = (3, 8)$ and $\eta = 0.5$, increasing r from zero to two, four and six reduced mean TV from 0.17565 to 0.05215, 0.00938 and 0.00066. At $\eta = 0.95$ the corresponding means were 0.35474, 0.25697, 0.18949 and 0.11176. Keeping all sectors gives zero error up to floating roundoff.

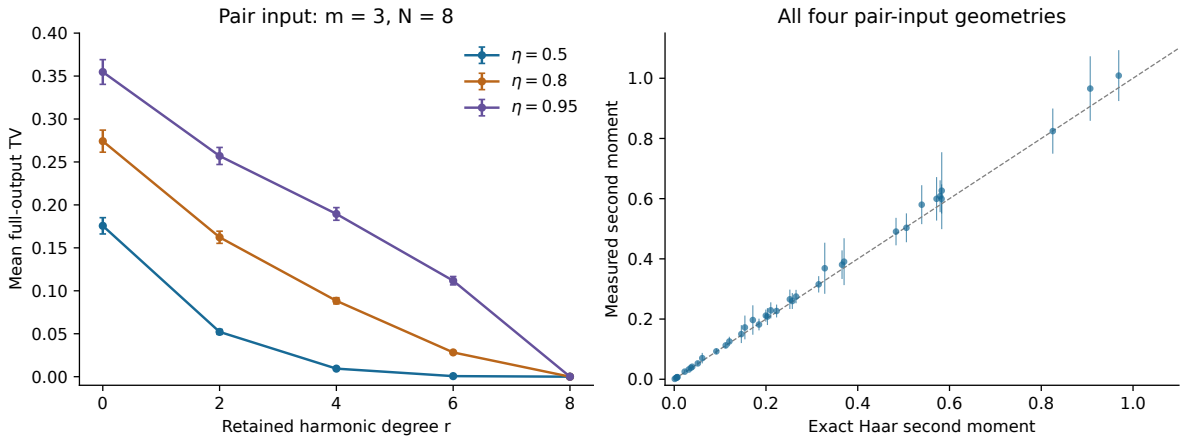


Figure 3: Conditional pair-state controls. Left: full-output mean TV after ideal harmonic truncation and proper sampling for $(m, N) = (3, 8)$. Bars show two pointwise standard errors over 128 Haar matrices. Right: exact Haar second-moment predictions against measured means across all four geometries; bars again show two standard errors. Parameter cells sharing an interferometer are correlated. These finite full-Fock computations are not runtime benchmarks.

11 Verification, reproducibility and scope

Version 2.0 assembles two complete theorem candidates, with proofs covering full photon-count outputs, Haar-average TV error and worst-case polynomial bit complexity at their stated fixed parameters. The Fock theorem answers the cited Open Problem Zoo formulation. The equal-squeezing Gaussian theorem is a separate extension. Additional vacuum inputs, unequal squeezing, nonuniform loss, arbitrary internal overlap matrices and a matching hardness boundary remain outside these results.

The accompanying package contains source, raw numerical data, mathematical checkers and a manifest of file hashes. The Fock checker uses exact rational real and imaginary parts, comparing a Leibniz determinant with separately enumerated permanents. It verifies 329 frame/factorial identities and inequalities and eight coefficient cases at degrees two through four. Deliberately removing spectators, a binomial denominator or heavy-mode weights is detected in four, seven and five cases, respectively.

The Gaussian checks compare several constructions:

- 990 exact reduction-spectrum controls and 28 direct finite-Fock reductions;
- 7,735 exact harmonic-weight ratio checks and 6,307 geometric-envelope checks, with closed norms compared to the reduction-purity recurrence;
- seven finite prefix cases at degrees two, four and six, detecting all four tested faults: determinant exponent, heavy-mode binomials, pair factorials and outer projection;
- three symbolic identities, 814 exact coherent-reduction comparisons and three further prefix comparisons, including degree eight, using a separately constructed reduction-kernel projector;

- nine exact finite-precision cases, all detecting the invalid assumption that raw rounded-matrix prefixes remain normalized and compatible.

The latter review exposed a missing small-photon precision argument in an intermediate Gaussian draft. Section 7.5 supplies an explicit Lipschitz bound and prefix-stability estimate for both branches. For example, rounding a two-mode rotation with entries $3/5, 4/5$ to $5/8, 6/8$ leaves a one-photon prefix weight $1/2$ but gives child sum $3721/8192$ under the raw two-photon formulas. The repaired proof controls this discrepancy instead of assuming it is zero.

Exact checks test finite algebraic instances; the prefix and Haar comparisons that use floating arithmetic are identified separately. None replaces the all-order analytic proof or benchmarks the asymptotic sampler. The targeted primary-source comparison establishes differences in the stated regimes of the checked papers, rather than global priority over every concurrent manuscript.

This work was developed with AI assistance under human direction, including derivation, literature comparison, computation, internal review and drafting. The separate computational constructions and proof audits belong to that same workflow and are not independent expert review. The release is an internally checked research manuscript; independent expert review remains outstanding.

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